Laboratory Studies

Unless the diagnosis is certain, important investigations for all photosensitive conditions include assessment of circulating anti-nuclear antibodies and extractable nuclear antibody titers, as well as measurement of blood, urine, and stool porphyrin concentrations.

Lesional histologic features are characteristic in many photodermatoses—particularly polymorphic light eruption (PMLE), hydroa vacciniforme (HV), and chronic actinic dermatitis (CAD)—but are rarely diagnostic except in HV. Direct immunofluorescence can assist in diagnosing cutaneous lupus.

Phototesting of normal back skin using a monochromator in CAD and solar urticaria (SU) often reproduces the papules or wheals typical of the condition, frequently at low irradiation doses, and helps define the action spectrum of the eruption. In xeroderma pigmentosum (XP), phototesting typically shows delayed erythema developing over 2 to 3 days with an abnormally low dose threshold, often progressing to blistering.

However, phototesting is only moderately reliable in other photodermatoses. Solar-simulating or broadband irradiation may still be useful to induce the rash for diagnostic purposes. In eczematous photosensitivity, patch and photopatch testing are essential to identify potential allergens that induce or exacerbate the condition.

Specialized techniques such as assessment of DNA excision repair or RNA synthesis recovery rate in cultured patient fibroblasts after ultraviolet radiation (UVR) exposure are required for diagnosing certain genophotodermatoses.

Phototesting

The use of phototesting in clinical dermatology varies widely between countries and centers. While it is the investigation of choice when the diagnosis of a photodermatosis is uncertain or when details of the action spectrum are needed, it remains primarily a research tool available at only a limited number of clinical centers.

Phototesting in suspected photodermatosis falls into two main categories: - **Monochromatic phototesting**: Performed on the upper back at a series of wavelengths and doses to elicit the eruption and identify the action spectrum. - **Photoprovocation testing**: Uses a broad-spectrum source to induce the eruption for clinical diagnosis or biopsy when necessary.

Monochromatic phototesting should ideally be performed with a xenon arc irradiation monochromator for precise determination of wavelength dependency. For photoprovocation, a solar simulator—typically a filtered xenon arc source that mimics terrestrial sunlight at noon on a midsummer day in temperate regions—is preferred.

Alternative protocols using broad-spectrum metal halide or fluorescent light sources with appropriate filters have been described. In some settings, filtered sunlight has been used, though this method is generally too variable for reliable clinical use.

Monochromatic testing is best conducted on unaffected skin of the upper back, lateral to the paravertebral groove. Lesion induction—except in conditions like SU and CAD where monochromators are effective—is more reliably achieved with broadband sources over larger areas of skin susceptible to the eruption. Conditions such as PMLE, actinic prurigo (AP), and HV often require repeated irradiation with UVA, UVB, or combined sources to reproduce the characteristic skin changes.

Precautions and Standardization

To avoid false-negative results, potent topical and systemic corticosteroids should be discontinued for several days before testing if possible. The impact of oral immunosuppressive agents on test results is less clear, but they should also ideally be withheld.

False-positive results may occur in patients with widespread active disease. Whenever possible, the eruption should be well-controlled before testing, which may involve keeping the patient in a low-light environment. However, these ideal conditions are often difficult to achieve during active disease, and testing may proceed with the understanding that results could be unreliable.

All phototesting must be performed under carefully standardized conditions, using sequential doses (often in a geometric series) and specific wavelengths. Results should be read at consistent time points after exposure, under controlled environmental conditions of light and temperature.